

Chapter 1 — Introduction to datasets

Define a dataset as curated records and attributes prepared for a purpose

Learning objectives

- Define what counts as a dataset (vs raw logs)
- Distinguish records (rows) from attributes (columns)
- Recognize the same idea in CSV and JSON forms

What is a dataset?

- Curated collection of data for analysis or decisions
- Typically organized as records and attributes
- Prepared for a purpose: scope, structure, documentation
- Basic unit for ML, statistics, and business intelligence

Records and attributes

- **Record** — one observation (row, instance)
- **Attribute** — one property of that observation
(column, field)
- Sales example: one sale = one record; Date, Product,
Price = attributes
- Same pattern in healthcare, sensors, and other
domains

Example 1.1 — Sample CSV sales data

```
$ cd modules/chapter1/example1/ && bash run.sh
```

Example 1.4 — Housing prices in JSON

```
$ cd modules/chapter1/example4/ && bash run.sh
```

Takeaways

- A dataset is curated for a purpose, not just a dump of bytes
- Records + attributes are the core vocabulary
- Form varies (CSV, JSON, images, text); the goal stays insight